Online Fault Diagnosis and Tolerant Control Based on Belief Rule Base Expert System Integrating Multisource Unreliable Information

Yijie Sun, Zhichao Feng, Zhijie Zhou, Changhua Hu, and Ruohan Yang

***Abstract*—This article develops a new online fault diagnosis and tolerant control framework for sensor failures based on belief rule base (BRB) expert system to solve three problems in engineering practice: lack of data in system failure state, expert knowledge uncertainty and complex environment interference. First of all, the new framework incorporates both quantitative information and qualitative knowledge, aiming to handle the influence of the lack of high-value failure data and uncertainty of expert knowledge. Second, this article proposes a new BRB model considering dynamic attribute reliability (BRB-DR) by contemplating the impact of complex environments on multisource information. In addition, the BRB-DR based online fault diagnosis and fault tolerant control (OFTC) framework is developed for handling sensor failures, where the model structure and parameters are both adjusted adaptively. Third, a sensitivity analysis of the dynamic attribute reliability to OFTC framework output is performed based on the iterative BRB-DR model, that can also provide decision making support for system improvement design. A case study based on wireless sensor network in rocket has been conducted to demonstrate the validity of the new framework.**

***Index Terms*—Belief rule base, fault diagnosis and tolerant control, multisource unreliable information, sensitivity analysis.**

I. Introduction

With the increasing of complexity and intelligence, observation information is becoming one of the most important input sources to provide the decision making of complex systems [1], [2], [3], [4]. Therefore, ensuring the reliability and accuracy of observation information is the crucial measure to improve performance of complex systems [5], [6], [7], [8].

Sensor networks are one of the core means to collect observation information from complex systems and can be used to generate control inputs for actuators. Once the sensor fails, its observation information will be inaccurate or lost [9]. This situation will reduce the precision of controller control input, and further weakening the performance of complex systems. Therefore, it is necessary to take a couple of measures to enhance the reliability and accuracy of sensor networks. Fault diagnosis and fault-tolerant control are important ways to improve the reliability of complex systems [10], [11], [12], [13], whose purpose is to detect and diagnose faults, and then takes fault tolerance strategies to handle the influence of the failure [14], [15], [16]. Currently, many excellent research have been carried out with the aim of improving the performance of sensor fault diagnosis and fault-tolerant control. For instance, Campos-Delgado et al. [17] innovated a new sensor failure-tolerant control method by robustness controller. Yong et al. [18] designed a fault tolerant control scheme to improve the reliability of voltage source inverts. Abdullah et al. [19] proposed a new fault tolerant control scheme for linear parameter varying system.

In practice, there are mainly three problems that need to be solved in sensors fault diagnosis and tolerant control. First, the sensor design is usually characterized by high reliability, resulting in a small amount of observed data during system failure states. However, as the cumulative work events increase, the performance of its material will degrade and the sensor failure is inevitable. For example, there are fewer fault samples for bearings, which are critical components in the aerospace field, leading to a lack of training samples and difficulty in building accurate fault diagnosis models [20]. In addition, in the sensor network of complex systems, sensors of same type are distributed differently, and the interference factors they receive are also different. Thus, experts cannot provide accurate judgement of the sensor status, while the cost of establishing an adequate mathematical model is high [13]. Second, in practical system, noise may be introduced into observation information for the influence of disturbances in actual environment, thus reducing its reliability [21]. Sensor state cannot be represented by the observation information accurately that further influences accuracy of the fault diagnosis and tolerant control once the sensor observation information is unreliable. In wireless sensor networks, external interference affects the operation of the sensors, thereby altering the sensing results and introducing biases in the values of the measured attributes [22]. Third, once the observation information of sensors is unreliable, measures should be taken to handle its influence. To minimize the impact of unreliable observation information and maximize the use of

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resources, it is necessary to give priority to the characteristics that have a large impact on the diagnosis and tolerant control output. Hence, the impact of the attribute reliability to fault diagnosis accuracy and tolerant control performance should both be quantitatively analyzed. For example, in fault-tolerant control of intelligent driving vehicles [23], failures and disturbances need to be taken into account to improve the reliability and performance of the system by designing adaptive control strategies. If the unreliable information collected by sensor *A* have a greater effect on the output than sensor *B*, it will have a higher effect when all resources are used to improve the reliability of sensor *A*. Thus, the above three problems need to be solved in the context of fault diagnosis and sensor fault-tolerant control of complex systems.

Belief rule base (BRB) is a kind of expert system, which is developed based on three theories: evidential reasoning theory, IF-THEN rule and Fuzzy theory [24], [25]. BRB not only inherits the fuzzy processing ability of the fuzzy system, but also further improves the processing ability of uncertainty and incomplete information by introducing the confidence structure and evidential reasoning algorithm [26]. The most advantageous feature of BRB is that it can integrate both quantitative data and qualitative knowledge simultaneously [27], [28], [29], [30], [31], [32]. What is more, the modeling process of BRB is traceable and transparent, where the evidential reasoning (ER) method is used to aggregate its belief rules. Thus, BRB has an excellent performance to deal with the lack of fuzzy expert knowledge and high-value observation data. For instance, Li et al. [33] developed a safety assessment model based on BRB and the conditional generalized minimum variance to address high-dimensional characteristics, Hu et al. [34] proposed a optimization method for BRB based on the method of feasible direction to ensure its traceability, Chang et al. [35] developed a structure learning method to reduce the model complexity. In the BRB model, relationship between inputs and outputs is expressed by belief rules [36], [37]. When the number of input features is larger, the complexity of BRB will be extremely higher, which will reduce the real-time performance of BRB model. At the same time, the amount of the belief rule has direct relationship with the model accuracy. Hence, to balance the model complexity and the accuracy at the same time, He et al. proposed a belief rule reliability based model reduction method, where the belief rule reliability is calculated by considering the static reliability of the attribute and the global ignorance [38]. However, for fault diagnosis and tolerant control of sensor failure for dynamic system, not only the method's real-time performance should be considered, but also the changes of the environment interference. Thus, the dynamic reliability of the attribute should be considered.

In this article, on the basis of the model reduction method by He et al. [38], an online fault diagnosis and tolerant control framework (OFTC) for sensor failure has been constructed for the first time, where BRB with dynamic attribute reliability (BRB-DR) is developed to handle the unreliable observation information. It is worth pointing out that the output reconstruction strategy is applied for failure sensor. Moreover, a dynamic reliability calculation method for attributes is proposed based on segment statistics strategy. Then, the structure of the OFTC can be adjusted based on the dynamic model reduction method with dynamic attribute reliability. To quantitatively analyze the its validity, the attribute reliability to the fault diagnostic outputs and the tolerance control outputs are sensitively analyzed based on the transparency and traceability of BRB-DR. The sensitivity coefficients of the attributes can be sorted by size and provide decision making support for system design. The initial OFTC framework is constructed by experts. For the reason that the initial OFTC framework may not accurately reconstruct the output of failure sensor due to the uncertainty in the knowledge of the experts, optimization model based on projected covariance matrix adaptive evolution strategy (P-CMA-ES) is developed for the parameters and structure of the OFTC framework in this article [38].

The rest of this article is organized as follows. In Section II, three problems in fault diagnosis and tolerant control of complex systems are formulated, and then the OFTC framework is developed. The BRB-DR is developed in Section III, while the inference process of the OFTC framework is carefully presented. Sensitivity analysis of the dynamic attribute reliability to fault diagnosis output and tolerant control output is carried out in Section IV. In Section V, a case study is conducted to justify the rationale for the model. Finally, Section VI concludes this article.

II. Problem Formulation and Framework Construction

This section first presents the problems in online diagnosis and fault-tolerant control of sensor faults, and then constructs a BRB-DR-based OFTC framework.

A. Problem Formulation

The current online fault diagnosis and fault-tolerant control of complex systems sensors suffers from the following three main problems.

Problem 1: For observation sensors in complex systems, which are generally highly reliable, failures are inevitable when they are working for a long time, which will cause fewer failure data in the observation data and fail to provide enough sensor information to allow the development of an accurate fault diagnosis and fault-tolerant control model. What is more, experts are unable to offer accurate knowledge about the state of sensors in complex systems, and the uncertainty, ambiguity and incompleteness of the knowledge also increase the difficulty in using it. Hence, the cost of developing accurate mathematical model is high. These two aspects are the first problem to be solved.

Problem 2: The discrete-time nonlinear system can be expressed as follows:

$$\begin{cases} \ell(t+1) = \Xi_1\{\ell(t), \lambda(t), c_1(t)\} \\ z(t) = \Xi_2\{\ell(t), c_2(t)\} \end{cases} \tag{1}$$

where $\ell(t)$ and $\ell(t+1)$ are the states of the sensor network for complex systems at time instant t and $(t+1)$. $\lambda(t)$ stands for

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the control input, while $z(t)$ represents the observation data from sensor network. $c_1(t)$ and $c_2(t)$, respectively, denote the system noise and the observation noise. Once a sensor failure occurs, accurate observations cannot be collected, and $z(t)$ becomes $z_F(t)$, which influences the accuracy of $\lambda(t)$ and further decreases the performance of the complex systems. Here, $z_F(t)$ stands for the failed monitoring data. For $\ell(t)$, not all states need to be observed or can be observed. Moreover, the dimension of $z(t)$ is determined by the system design. Hence, for the dimension, there is no relationship between the $\ell(t)$, $z(t)$ and $\lambda(t)$.

In practice, affected by the disturbance factors of the sensor environment, their observation data contain noise that reduce their signal-to-noise ratio. These unreliable observation data are unable to express the accurate sensor state information. In addition, for dynamic complex systems, the environment disturbance factors will change, and then the noise in the observation data also changes. Therefore, the following framework needs to be constructed to reconstruct the unreliable input information $z(t)$ in order to maintain the state of the system

$$\{\hat{z}(t), F(t)\} = \Psi\{z(t), z_F(t), r(t), \mathbf{R}\} \tag{2}$$

where $\hat{z}(t)$ is the observation from reconstructed faulty sensor. $F(t)$ denotes the diagnosis output of OFTC framework. $r(t) = \{r_1(t), \ldots, r_M(t)\}$, and it represents the dynamic reliability of the M sensors. $\mathbf{R}$ stands for the expert knowledge applied to develop the OFTC framework.

Problem 3: The unreliable observation information of sensors directly influences the OFTC framework's accuracy, and it is determined by the performance of sensors. For practical system, the sources that can be used are limited. It would be unrealistic to enhance the reliability of all sensors. However, if the effectiveness of the unreliability of sensors are able to be calculated, they can be sorted to provide guidance for users to improve system design. Thus, how to analyze the effectiveness of the unreliable observation information gathered by sensors is the third problem.

B. OFTC Framework Construction

Aiming at the above three problems, this article considers the dynamic reliability of sensor observation information and establishes the BRB-DR model. In the BRB-DR model, the dynamic reliabilities of the input attributes have been considered, and multiple belief rules coexist at the same time. The kth belief rule can be described as follows [39], [40]:

$$\begin{aligned} E_k(t): \text{If } & b_1(t) \text{ is } V_1^k \wedge b_2(t) \text{ is } V_2^k \cdots \wedge b_M(t) \text{ is } V_M^k, \\ & \text{then } B(t) \text{ is } \{(\phi_1, p_{1,k}), (\phi_2, p_{2,k}), \ldots, (\phi_N, p_{N,k}), \\ & (\Theta, p_{\Theta,k})\}, \left(\sum_{n=1}^{N} p_{n,k} \le 1, \sum_{n=1}^{N} p_n + p_\Theta = 1\right) \\ & \text{with attribute weight } \lambda_1(t), \cdots, \lambda_M(t), \text{rule weight } \delta_k(t), \\ & \text{and attribute reliability } r_1(t), r_2(t), \ldots, r_M(t) \end{aligned} \tag{3}$$

where $b_1(t), b_2(t), \ldots, b_M(t)$ are the M input attributes of BRB-DR. $V_1^k, V_2^k, \ldots, V_M^k$ are their corresponding reference which has the purpose to transform multiple sources of data under a unified framework. $\phi_1, \phi_2, \ldots, \phi_N$ denote the output values of the BRB-DR model, and $p_{1,k}, p_{2,k}, \ldots, p_{N,k}$ represent their belief degrees in the kth belief rule. Note that the sum of the output belief degree $p_{n,k}$ is less than 1 influenced by the incomplete input information. $p_{\Theta,k}$ is the remaining degree of belief, and it is not possible to distinguish which rank should be exported. $\delta_k(t)$ stands for the rule weight of the kth belief rule. $\lambda_1(t), \lambda_2(t), \ldots, \lambda_M(t)$ and $r_1(t), r_2(t), \ldots, r_M(t)$ represent the weight and reliability of the M attributes at time t, respectively.

Remark 1: Each input attribute of the model corresponds to a monitoring sensor of the complex system. Attribute weight belongs to the external parameter of an attribute and indicates the relative importance between sensors. Reliability is an internal parameter of an attribute, indicating the degree of confidence in the data collected by the sensor corresponding to the attribute, which is related to the health status of the sensors themselves, their length of service, and the degree of interference in the working environment. Weight and reliability are independent and do not affect each other.

Then, the OFTC framework can be established by the developed BRB-DR. In this article, the OFTC framework consists of two parts: 1) a fault diagnosis model and 2) a sensor fault output reconstruction model. In the BRB-DR based fault diagnosis model, the sensor observations are taken as inputs while the outputs are faulty sensors. Furthermore, observations from available sensors are used as inputs to the output reconstruction model, and estimated observations from faulty sensors are its outputs.

III. OFTC Framework Inference Process

In this section, a calculation method of dynamic reliability for attribute in complex systems is proposed in Section III-A, and then the inference process of the BRB-DR model is presented in Section III-B. Section III-C describes the fault diagnosis and fault-tolerant control framework. In order to deal with the effect of uncertain expert knowledge of Problem 1, a new optimization model for the OFTC framework is developed in Section III-D.

A. Calculation of Dynamic Reliability

In order to deal with the effects of unreliable information in Problem 2, this part proposes a dynamic reliability calculation method. For dynamic system, its state changes timely, and can mainly be divided into the following three categories: Steady state, slowly changing state, and step state. Therefore, there is a need for a reliability calculation method that is updated online in real time. The calculation method of dynamic attribute reliability is proposed by the staged statistics-based method.

1) *Case 1. Steady state:* When the system state is stable, the changes of the observation data are only affected by environmental interference. When there is no environmental noise, the observation data of sensors should be a constant value. On the contrary, the observation data influenced by the environment interference will fluctuate [34]. The

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environment noise will cause the observation data to fluctuate up and down. Thus, within the time window ϖ, the attribute reliability of complex systems gathered by the mth sensor is calculated as follows:

$$r_m(t) = \frac{T - y_m(t)}{T}, \tag{4}$$

$$y_m(t) = \sum_{j=1}^{T} y_{mj} \tag{5}$$

where $r_m(t)$ is the reliability gathered by the mth sensor at time t. T indicates the amount of observed data in the time window ϖ. Note that the time window represents the observation data from current moment to the previous $(t-T+1)$th historical moment. It is up to the expert to determine the length of the time window based on the operating environment and the state of the system. $y_m(t)$ denotes the number of the reliable observation information of the mth system attribute at current time. y_{mj} represents the observation data in the time window ϖ. The reliable observation information are selected by the fluctuation range that can be profiled as $(\overline{y}_m - \tau\sigma_m, \overline{y}_m + \tau\sigma_m)$, where $\overline{y}_m$ and σ_m are the mean and variance of the ith system attribute. τ aims to adjust the width of the fluctuation range. When the jth observation is outside the fluctuation range, $y_{mj} = 1$; otherwise, $y_{mj} = 0$.

2) *Case 2. Slowly changing state:* When the state of the system varies slowly, the observation information will vary regularly, such as performance degradation. With a period of time, the change trend of the observation data can be approximated as a linear change. The trend of the system state could be analyzed using the following equation:

$$y_{mj} = \upsilon_1 \varphi_j + \upsilon_2 \tag{6}$$

where $j = (t-T+1), \ldots, t$. υ_1 and υ_2 represent the parameters in the linear equation that could be obtained by maximum likelihood estimation according to the following equations:

$$\hat{\upsilon}_1 = \frac{\nu \sum_{j=t-T+1}^{t_i} \Delta\varphi_j}{\sum_{j=t-T+1}^{t_j} (\Delta\varphi_j)^2} \tag{7}$$

$$\hat{\upsilon}_2 = \bar{y} - \hat{\upsilon}_1 \bar{\varphi}. \tag{8}$$

In this article, assume that the change of the system is a Markov process, then $\Delta y_{mj} = y_{mj} - y_{m(j-1)}$ and $\Delta y_{mj} \sim N(\nu, \epsilon^2)$. After that, the fluctuation range can be expressed as $(y_{mj} - \tau\epsilon, y_{mj} + \tau\epsilon)$. That is to say, the fluctuation range changes according to the system state. The unreliable observation information are available in terms of fluctuation range. The reliability of the system attribute gathered by the mth sensor can be calculated by (4) and (5).

3) *Case 3. Step state:* When the system state occurs a step change, the observation data fluctuations greatly. In this case, the large fluctuation of the observation data is caused by the change of system state, not caused by the noise. Hence, if we use the statistics based method to calculate the reliability of system attribute, the gathered result is unreasonable. For example, when the rocket is ready to early fly, the observation information of the shock sensor changes greatly while environment interference has not changed.

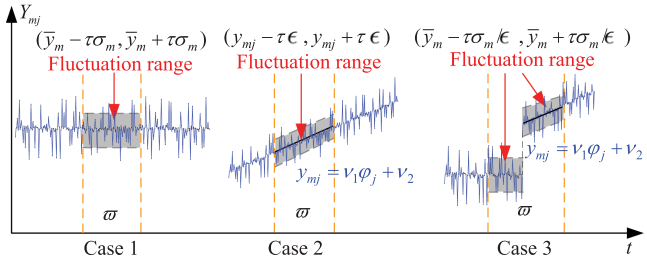


Fig. 1. Three cases of dynamic attribute reliability

For the step state of system, it needs to determine the time to step. Thus, the step judgement threshold $\hbar$ should be determined firstly. If $y_{mj} > \hbar$, the system state has a step change at time instant j. The state change of the two phases of the system may be constant or linear, and the calculation method developed in *Case 1* and *Case 2* can be applied to calculate the attribute reliability of two phases, respectively. It should be noted that the different reliability should be used in different phases.

To clearly present the calculation procedure of dynamic attribute reliability, a numerical case is provided as shown in Fig. 1, where three cases exist. In Fig. 1, the blue lines represent the range of fluctuations in the data observations. The gray area represents the data reliability area, while the area beyond the gray data will be considered unreliable and is divided into three cases to calculate its unreliability.

B. Inference Process of BRB-DR

The following three steps exist in the reasoning process of the BRB-DR model.

1) *Step 1. Observation data transformation:* In engineering practice, the observation data of complex systems collected by sensors are in different formats and cannot be directly fused due to the existence of multisource attributes. Therefore, multisource information should be converted into a unified format [41]. The conversion equation is as follows:

$$\alpha_j^i(t) = \begin{cases} \frac{V_i^{l+1} - b_i(t)}{V_i^{l+1} - V_i^l}, & j = l, \text{ if } V_i^l \le b_i(t) \le V_i^{l+1} \\ \frac{b_i(t) - V_i^l}{V_i^{l+1} - V_i^l}, & j = l+1 \\ 0, & j = 1, 2, \ldots, L, j \ne l, l+1 \end{cases} \tag{9}$$

where $\alpha_j^i(t)$ denotes the match degree of the ith input in the jth rule at time t. $b_i(t)$ is the observation of the ith input at time t. V_i^l represents the ith reference value.

It is worth noting that the belief rules need to satisfy that the reference value of any observation input of the next

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rule is greater than or equal to the previous rule when applying (9). When assigning the matching degree, the same matching degree is assigned to the same reference value of the same observation input. Thus, the first step aims to transform the multisource observation data into the matching degree that can be expressed as follows:

$$(b_1(t), b_2(t), \cdots, b_M(t)) \Rightarrow$$
$$(V_1^l, \alpha_l^1(t)) \wedge (V_2^l, \alpha_l^2(t)) \wedge \cdots \wedge (V_M^l, \alpha_l^M(t)) \quad (10)$$

where M is the number of the input attribute in BRB-DR.

2) *Step 2. Rule activation weight calculation:* The belief rules in the BRB-DR model are then activated to varying degrees by the input observations collected by the sensor. The match between the feature input and the kth belief rule can be calculated by the following equations:

$$\alpha_k(t) = \prod_{i=1}^{M} (\alpha_k^i(t))^{\chi_i(t)} \quad (11)$$

$$\chi_i(t) = \frac{\overline{\delta}_i(t)}{(1 + \overline{\delta}_i(t) - r_i(t))} \quad (12)$$

$$\overline{\lambda}_i(t) = \frac{\lambda_i(t)}{\max\limits_{i=1,\ldots,T_k}\{\lambda_i(t)\}}, 0 \le \overline{\lambda}_i(t) \le 1 \quad (13)$$

where $\alpha_k(t)$ represents the total matching degree of M feature inputs for the kth rule at moment t. $\chi_i(t)$ denotes the utility of ith feature that considers its reliability and weight simultaneously. $\overline{\lambda}_i(t)$ is the relative weight of the ith feature.

Then, the activation weights of the belief rules in BRB-DR can be found by the following equation:

$$w_k(t) = \frac{\theta_k(t)\alpha_k(t)}{\sum_{l=1}^{L} \theta_l(t)\alpha_l(t)} \quad (14)$$

where $w_k(t)$ $\theta_k(t)$ denotes the activation weight and the weight of the kth belief rule at time t, respectively.

3) *Step 3. Belief rule convergence:* In this article, the belief rules are integrated after activation through ER method. Instead of precise a priori probability and conditional probability, it expresses uncertainty through the basic probability distribution function, which can simultaneously deal with multiple uncertainties such as randomness, ambiguity, inaccuracy and inconsistency, and can effectively deal with the conflict between the evidence.

The basic probability mass (*bpm*) is obtained from the following equations [42]:

$$S_{n,k} = w_k p_{n,k}, \ n = 1, \cdots N \quad (15)$$

$$S_{R,k} = 1 - \sum_{n=1}^{N} S_{n,k} = 1 - w_k \sum_{n=1}^{N} p_{n,k} \quad (16)$$

$$\overline{S}_{R,k} = 1 - w_k \quad (17)$$

$$\widetilde{S}_{R,k} = w_k \left(1 - \sum_{n=1}^{N} p_{n,k}\right) \quad (18)$$

where $S_{n,k}$ denotes the basic probability mass in the kth belief rule. $S_{R,k}$ represents the remaining basic probability mass. Note that $S_{R,k} = \overline{S}_{R,k} + \widetilde{S}_{R,k}$. $\overline{S}_{R,k}$ stands for the underlying probabilistic quality of the residuals given to the other rules and is caused by the activation weights. $\widetilde{S}_{R,k}$ denotes the uncertain underlying probability mass assigned to the kth rule, caused by the incompleteness of the output of the confidence rule.

Then, the activated belief rules can be successively integrated by

$$S_{n,e(k)} = K_{e(k)}[S_{n,e(k-1)}S_{n,k} + S_{n,e(k-1)}S_{R,k} + S_{R,e(k-1)}S_{n,k}] \quad (19)$$

$$S_{R,e(k)} = \bar{S}_{R,e(k)} + \tilde{S}_{R,e(k)} \quad (20)$$

$$\tilde{S}_{R,e(k)} = K_{e(k)}[\tilde{S}_{R,e(k-1)}\tilde{S}_{R,k} + \tilde{S}_{R,e(k-1)}\bar{S}_{R,k} + \bar{S}_{R,e(k-1)}\tilde{S}_{R,k}] \quad (21)$$

$$\bar{S}_{R,e(k)} = K_{e(k)}[\bar{S}_{R,e(k-1)}\bar{S}_{R,k}] \quad (22)$$

$$K_{e(k)} = \left[1 - \sum_{n=1}^{N}\sum_{\substack{t=1 \\ t\neq n}}^{N} S_{n,e(k-1)}S_{t,k}\right]^{-1} \quad k = 2, 3, \ldots, L \quad (23)$$

where $S_{n,e(k)}$ is the *bpm* of the nth output degree after the k belief rules are aggregated. $S_{R,e(k)}$ represents the residual *bpm*. $K_{e(k)}$ is the normalizing factor that aims to ensure $\sum_{n=1}^{N} S_{n,e(L)} + S_{R,e(L)} = 1$.

The aggregated belief degree of the BRB-DR could be obtained as follows:

$$p_n = \frac{S_{n,e(L)}}{1 - \bar{S}_{R,e(L)}} \quad (24)$$

$$p_D = \frac{\tilde{S}_{n,e(L)}}{1 - \bar{S}_{R,e(L)}} \quad (25)$$

where p_n and p_D denote the probability that the output is the result n or D, respectively. In the BRB-DR model, $\sum_{n=1}^{N} p_n + p_D = 1$.

C. Fault Diagnosis and Output Reconstruction Strategy

According to the BRB-DR model, the multisource observation information gathered by sensors can be combined and its final outputs are represented by (24) and (25). In the OFTC framework, the nearest neighbor strategy and the output utility strategy are used in the fault diagnosis and the output reconstruction model (FDOR), respectively.

In the fault diagnosis model, the standard failure variables of sensors should be given as

$$[\mathbf{V_1}, \mathbf{V_2}, \cdots \mathbf{V_N}]^{\mathrm{T}} = \begin{bmatrix} v_{11} & \cdots & v_{1N} \\ \vdots & \ddots & \vdots \\ v_{N1} & \cdots & v_{NN} \end{bmatrix} \quad (26)$$

where $\mathbf{V}_i$ denotes the eigenvector of ith fault class, $\mathbf{V}_i = [\mathrm{v}_{1i}, \mathrm{v}_{2i} \ldots \mathrm{v}_{Ni}]^{\mathrm{T}}$. It is important to notice that the standard

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failure variables of the different sensors are different. Then, the distance between the BRB-DR output and the standard failure variable can be calculated as follows:

$$\|\mathbf{p}(t)-\mathbf{V}_n\|=\left(\sum_{i=1}^{N}|p_i(t)-v_{ni}|^2\right)^{1/2} \tag{27}$$

where $\mathbf{p}(t)=[p_1(t),p_2(t),\ldots,p_N(t)]^{\mathrm{T}}$. Note that the two-norm is applied to calculate the distance in the above formula, and other norms can also be used according to the practical system.

Then, the failure sensor is able to be diagnosed by the distance between the standard failure variable and the BRB-DR model output that can be shown by

$$\|\mathbf{p}(t)-\mathbf{V}_l\|=\min_{1\le n\le N}\|\mathbf{p}(t)-\mathbf{V}_n\|. \tag{28}$$

That is to say, as shown in the above equation, the lth sensor fails at time t. Then, after the failure sensor has been diagnosed, its output in terms of observation information from other available sensors can be reconstructed. In this article, the observation information of the sensor is calculated as follows:

$$\hat{z}_l(t)=\sum_{n=1}^{N}p_n(t)q_{ln} \tag{29}$$

where $\hat{z}_l(t)$ represents the reconstructed output of the lth failure sensor. q_{ln} is the utility of the nth output degree of the lth failure sensor, and it is applied to reconstruct the failure sensor output.

Remark 2: In practice, the quality of various sensors is different, and the degradation speed of sensors at different locations is also different. Hence, in the BRB-DR based OFTC framework as profiled, the amount of the input attributes M may not equal to the number of the failure cases N in engineering practice.

D. Optimization Model

The initial OFTC framework is unable to maintain the performance of the system under sensor failure because of the uncertain expert knowledge. Moreover, the construction of the initial OFTC framework needs to follow the principle of ergodicity. However, for practical system, influenced by the environment noise, some belief rules do not work or are less effective. These belief rules will make the framework more complex while reduce its real-time performance. Therefore, in this subsection, to cope with the effects of uncertain expert knowledge and the useless belief rule, an optimization model of the OFTC framework is proposed based on the P-CMA-ES method [13].

In the optimization model of the OFTC framework, there are two main parts: 1) parameter optimization and 2) structural optimization.

1) *Parameter optimization:* The optimization target is to minimize the error between the actual output and the model output. The mean square error (MSE) can serve as an indication of the precision of the framework. The objective functions [40] of the FDOR model are calculated by

$$\min \mathrm{MSE}_{FD}=\min\frac{1}{T_{FD}}\sum_{t=1}^{T_{FD}}\sum_{n=1}^{N}(p_n(t)-C_n(t))^2 \tag{30}$$

$$\min \mathrm{MSE}_{RE}=\min\frac{1}{T_{RE}}\sum_{t=1}^{T_{RE}}(\hat{z}_l(t)-\tilde{z}_l(t))^2 \tag{31}$$

where $C_n(t)$ and $\tilde{z}_l(t)$ stand for the actual outputs of FDOR model at time t, respectively, while T_{FD} and T_{RE} are their corresponding the observation data number.
The parameters in the OFTC framework should satisfy the actual physical meaning and comply with the constraints as shown as follows:

$$0\le p_{n,k}^{FD}(t),p_{n,k}^{RE}(t)\le 1 \tag{32}$$

$$0\le \delta_k^{FD}(t),\delta_k^{RE}(t)\le 1 \tag{33}$$

$$0\le \lambda_i^{FD}(t),\lambda_i^{RE}(t)\le 1 \tag{34}$$

$$0\le \sum_n p_{n,k}^{FD}(t),\sum_e p_{n,k}^{RE}(t)\le 1 \tag{35}$$

where $p_{n,k}^{FD}(t)$ and $p_{n,k}^{FD}(t)$ represent the output possibility in the kth belief rule, respectively. $\delta_k^{FD}(t)$, $\delta_k^{RE}(t)$, $\lambda_i^{FD}(t)$, and λ_i^{RE} are their corresponding rule and attribute weights.

2) *Structure optimization:* In the OFTC framework, as soon as the input attribute become unreliable, the belief rule constructed by the input attributes will also become unreliable. These unreliable belief rules cannot provide accurate relationship between input attributes and output degrees. In other words, the outputs of these belief rules are unreliable. Thus, the framework structure should be adjusted adaptively according to the rule reliability. The belief rule reliability can be expressed by

$$R_k(t)=r_1(t)r_2(t)\cdots r_M(t)\left(1-\frac{p_{\Theta,k}(t)}{\wp}\right) \tag{36}$$

where $R_k(t)$ is the reliability of the kth belief rule at time t. $p_{\Theta,k}(t)$ represents the remaining belief degree caused by the unreliable and uncertain expert knowledge. $\wp$ is an adjustment factor, and it aims to ensure the belief rule reliability satisfies the constraint $0\le R_k(t)\le 1$.

Once the belief rule reliability cannot meet requirement, it should be reduced to improve the accuracy of the OFTC framework. Thus, the framework structure can be adjusted by

$$\hat{\delta}_k(t)=\begin{cases}0, & R_k(t)\le R_{\text{threshold}}\\ \delta_k(t), & R_k(t)>R_{\text{threshold}}\end{cases} \tag{37}$$

where $\hat{\delta}_k(t)$ is the optimized rule weight. $R_{\text{threshold}}$ represents the rule reduction threshold given by experts. It can determine the balance between the complexity and accuracy of the framework.

IV. Sensitivity Analysis of Reliability

When the input observation information is disturbed by the environment noise, the output of the OFTC framework will be

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influenced. In practice, to handle the effects of the environment noise, some measures should be taken, such as the use of sensors with strong anti-interference ability. However, the use of these high-performance sensor will increase the cost of system design. Therefore, in regard to Problem 3, a quantitative analysis of the validity of the reliability of the dynamic attributes is needed in order to maximize the limited resources available in the real system. In this section, sensitivity analysis of the dynamic reliability of the BRB-DR output with respect to sensor observations is conducted [34].

Based on the (24), the first order derivative matrix of the BRB-DR output $p_n(t)$ to the ith attribute reliability $r_i(t)$ can be represented as follows:

$$\frac{\partial p_n(t)}{\partial r_i(t)} = \begin{bmatrix} \frac{1}{1-\bar{S}_{R,e(L)}} & \frac{S_{n,e(L)}}{\left(1-\bar{S}_{R,e(L)}\right)^2} \end{bmatrix} \begin{bmatrix} \frac{\partial S_{n,e(L)}}{\partial r_i(t)} \\ \frac{\partial \bar{S}_{R,e(L)}}{\partial r_i(t)} \end{bmatrix} \tag{38}$$

where $\frac{\partial S_{n,e(L)}}{\partial r_i(t)}$ and $\frac{\partial \bar{S}_{R,e(L)}}{\partial r_i(t)}$ can be obtained by

$$\frac{\partial S_{n,e(L)}}{\partial r_i(t)} = \frac{\partial S_{n,e(L)}}{\partial \mathbf{M}_L^{\mathrm{T}}} \frac{\partial \mathbf{M}_L}{\partial r_i(t)} \tag{39}$$

$$\frac{\partial \bar{S}_{R,e(L)}}{\partial r_i(t)} = \frac{\partial \bar{S}_{R,e(L)}}{\partial \mathbf{N}_L^{\mathrm{T}}} \frac{\partial \mathbf{N}_L}{\partial r_i(t)} \tag{40}$$

where $\mathbf{M}_L^{\mathrm{T}} = [K_{er(L)} \; S_{n,e(L-1)} S_{n,L} \; S_{R,L} \; S_{R,e(L-1)}]^{\mathrm{T}}$ and $\mathbf{N}_L^{\mathrm{T}} = [K_{er(L)} \; \bar{S}_{R,e(L-1)} \; \bar{S}_{R,L}]^{\mathrm{T}}$. It should be noted that $K_{e(L)}$, $S_{n,e(L-1)}$, $S_{n,L}$, $S_{R,L}$ and $S_{R,e(L-1)}$ are the final iteration parameters of ER method. In the combination process of the ER method, these parameters can be represented as $K_{e(l)}$, $S_{n,e(l-1)}$, $S_{n,k}$, $S_{R,l}$ and $S_{R,e(l-1)}$, respectively. Then, $\mathbf{M}_L^{\mathrm{T}}$ and $\mathbf{N}_L^{\mathrm{T}}$ can be represented by $\mathbf{M}^{\mathrm{T}} = [K_{e(l)} \; S_{n,e(l-1)} \; S_{n,l} \; S_{R,l} \; S_{R,e(l-1)}]^{\mathrm{T}}$ and $\mathbf{N}_L^{\mathrm{T}} = [K_{e(l)} \; \bar{S}_{R,e(l-1)} \; \bar{S}_{R,l}]^{\mathrm{T}}$.

In the (38), $\frac{\partial S_{n,e(l)}}{\partial r_i(t)}$ can be expressed by

$$\begin{aligned} \frac{\partial S_{n,e(l)}}{\partial \mathbf{M}^{\mathrm{T}}} &= \begin{bmatrix} \frac{\partial S_{n,e(l)}}{\partial K_{e(l)}} \frac{\partial S_{n,e(l)}}{\partial S_{n,e(l-1)}} & \frac{\partial S_{n,e(l)}}{\partial S_{n,l}} \frac{\partial S_{n,e(l)}}{\partial S_{R,l}} \frac{\partial S_{n,e(l)}}{\partial S_{R,e(l-1)}} \end{bmatrix} \\ &= \begin{bmatrix} S_{n,e(l-1)} S_{n,l} + S_{n,e(l-1)} S_{R,l} + S_{R,e(l-1)} S_{n,l} \\ K_{e(l)} \left(S_{n,l} + S_{R,l}\right) \\ K_{e(l)} \left(S_{n,e(l-1)} + S_{R,e(l-1)}\right) \\ K_{e(l)} S_{n,e(l-1)} \\ K_{e(l)} S_{n,l} \end{bmatrix}^{\mathrm{T}} \end{aligned} \tag{41}$$

$$\frac{\partial \mathbf{M}}{\partial r_i(t)} = \begin{bmatrix} \frac{\partial K_{e(l)}}{\partial r_i(t)} \frac{\partial S_{n,e(l-1)}}{\partial r_i(t)} & \frac{\partial S_{n,l}}{\partial r_i(t)} \frac{\partial S_{R,l}}{\partial r_i(t)} \frac{\partial S_{R,e(l-1)}}{\partial r_i(t)} \end{bmatrix}^{\mathrm{T}} \tag{42}$$

where $\frac{\partial S_{n,e(l-1)}}{\partial r_i(t)}$ is similar with $\frac{\partial S_{n,e(l)}}{\partial r_i(t)}$ that can be obtained by (38). $\frac{\partial K_{e(l)}}{\partial r_i(t)}$ can be expressed by

$$\frac{\partial K_{e(l)}}{\partial r_i(t)} = \begin{bmatrix} \frac{\partial \mathbf{K}}{\partial \mathbf{S}_{er}} & \frac{\partial \mathbf{K}}{\partial \mathbf{S}_t} \end{bmatrix} \begin{bmatrix} \frac{\partial \mathbf{S}_{er}}{\partial \mathbf{r}} \\ \frac{\partial \mathbf{S}_t}{\partial \mathbf{r}} \end{bmatrix} \tag{43}$$

with

$$\frac{\partial \mathbf{K}}{\partial \mathbf{S}_{er}} = \begin{bmatrix} \frac{\partial K_{e(l)}}{\partial S_{1,e(l-1)}} & \cdots & \frac{\partial K_{e(l)}}{\partial S_{N,e(l-1)}} \end{bmatrix} \tag{44}$$

$$\frac{\partial \mathbf{K}}{\partial \mathbf{S}_t} = \begin{bmatrix} \frac{\partial K_{e(l)}}{\partial S_{1,l}\atop t\neq n} & \cdots & \frac{\partial K_{e(l)}}{\partial S_{N,l}\atop t\neq n} \end{bmatrix} \tag{45}$$

$$\frac{\partial \mathbf{S}_{er}}{\partial \mathbf{r}} = \begin{bmatrix} \frac{\partial S_{1,e(l-1)}}{\partial r_i(t)} & \cdots & \frac{\partial S_{N,e(l-1)}}{\partial r_i(t)} \end{bmatrix}^{\mathrm{T}} \tag{46}$$

$$\frac{\partial \mathbf{S}_t}{\partial \mathbf{r}} = \begin{bmatrix} \frac{\partial S_{1,l}}{\partial r_i(t)}\atop t\neq n & \cdots & \frac{\partial S_{N,l}}{\partial r_i(t)}\atop t\neq n \end{bmatrix}^{\mathrm{T}} \tag{47}$$

$$\frac{\partial K_{e(l)}}{\partial S_{n,e(l-1)}} = \frac{\sum_{t=1\atop t\neq n}^{N} S_{t,l}}{\left(1 - \sum_{n=1}^{N} \sum_{t=1\atop t\neq n}^{N} S_{n,e(l-1)} S_{t,l}\right)^2} \tag{48}$$

$$\frac{\partial K_{e(l)}}{\partial S_{t,l}\atop t\neq n} = \frac{\sum_{n=1\atop n\neq t}^{N} S_{n,e(l-1)}}{\left(1 - \sum_{n=1}^{N} \sum_{t=1\atop t\neq n}^{N} S_{n,e(l-1)} S_{t,l}\right)^2} \tag{49}$$

where $\frac{\partial S_{n,e(l-1)}}{\partial r_i(t)}$ and $\frac{\partial S_{t,l}{}_{t\neq n}}{\partial r_i(t)}$ are similar with the $\frac{\partial S_{n,e(l)}}{\partial r_i(t)}$ and $\frac{\partial S_{n,l}}{\partial r_i(t)}$. $\frac{\partial S_{n,l}}{\partial r_i(t)}$ and $\frac{\partial S_{R,l}}{\partial r_i(t)}$ can be expressed as follows:

$$\frac{\partial S_{n,l}}{\partial r_i(t)} = p_{n,l} \frac{\partial w_l}{\partial r_i(t)} \tag{50}$$

$$\frac{\partial S_{R,l}}{\partial r_i(t)} = -\sum_{n=1}^{N} p_{n,l} \frac{\partial w_l}{\partial r_i(t)}. \tag{51}$$

In (42), the matrix of first order derivative of $S_{R,e(l)}$ to $r_i(t)$ can be obtained by the following equation:

$$\frac{\partial S_{R,e(l)}}{\partial r_i(t)} = \frac{\partial \bar{S}_{R,e(l)}}{\partial r_i(t)} + \frac{\partial \tilde{S}_{R,e(l)}}{\partial r_i(t)} \tag{52}$$

with

$$\frac{\partial \bar{S}_{R,e(l)}}{\partial r_i(t)} = \mathbf{A} \cdot \mathbf{B} \tag{53}$$

$$\frac{\partial \tilde{S}_{R,e(l)}}{\partial r_i(t)} = \mathbf{C} \cdot \mathbf{D} \tag{54}$$

where the first order derivation matrix of $\bar{S}_{R,e(l)}$ to $r_i(t)$ is the result of the (24). Then, $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, and $\mathbf{D}$ can be shown as

$$\begin{aligned} \mathbf{A} &= \begin{bmatrix} \frac{\partial \bar{S}_{R,e(l)}}{\partial K_{e(l)}} & \frac{\partial \bar{m}_{R,e(l)}}{\partial \bar{S}_{R,e(l-1)}} & \frac{\partial \bar{S}_{R,e(l)}}{\partial \bar{S}_{R,l}} \end{bmatrix} \\ &= \begin{bmatrix} \bar{S}_{R,e(l-1)} \bar{S}_{R,l} \; cK_{e(l)} \bar{S}_{R,l} \; K_{e(l)} \bar{S}_{R,e(l-1)} \end{bmatrix} \end{aligned} \tag{55}$$

$$\mathbf{B} = \begin{bmatrix} \frac{\partial K_{e(l)}}{\partial r_i(t)} & \frac{\partial \bar{S}_{R,e(l-1)}}{\partial r_i(t)} & \frac{\partial \bar{S}_{R,l}}{\partial r_i(t)} \end{bmatrix}^{\mathrm{T}} \tag{56}$$

$$\begin{aligned} \mathbf{C} &= \begin{bmatrix} \frac{\partial \tilde{S}_{R,e(l)}}{\partial K_{e(l)}} & \frac{\partial \tilde{S}_{R,e(l)}}{\partial \bar{S}_{R,e(l-1)}} \frac{\partial \tilde{S}_{R,e(l)}}{\partial \bar{S}_{R,l}} & \frac{\partial \tilde{S}_{R,e(l)}}{\partial \bar{S}_{R,l}} & \frac{\partial \tilde{S}_{R,e(l)}}{\partial \bar{S}_{R,e(l-1)}} \end{bmatrix} \\ &= \begin{bmatrix} \tilde{S}_{R,e(l-1)} \tilde{S}_{R,l} + \tilde{S}_{R,e(l-1)} \bar{S}_{R,l} + \bar{S}_{R,e(l-1)} \tilde{S}_{R,l} \\ K_{e(l)} \left(\tilde{S}_{R,l} + \bar{S}_{R,l}\right) \\ K_{e(l)} \left(\tilde{S}_{R,e(l-1)} + \bar{S}_{R,e(l-1)}\right) \\ K_{e(l)} \tilde{S}_{R,e(l-1)} \\ K_{e(l)} \tilde{S}_{R,l} \end{bmatrix}^{\mathrm{T}} \end{aligned} \tag{57}$$

$$\mathbf{D} = \begin{bmatrix} \frac{\partial K_{er}(l)}{\partial r_i(t)} & \frac{\partial \tilde{S}_{R,e(l-1)}}{\partial r_i(t)} & \frac{\partial \tilde{S}_{R,l}}{\partial r_i(t)} & \frac{\partial \bar{S}_{R,l}}{\partial r_i(t)} & \frac{\partial \bar{S}_{R,e(l-1)}}{\partial r_i(t)} \end{bmatrix}^{\mathrm{T}} \tag{58}$$

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where $\frac{\partial K_{er}(l)}{\partial r_i(t)}$ can be obtained by (42). The $\frac{\partial \bar{S}_{R,l}}{\partial r_i(t)}$ and $\frac{\partial \tilde{S}_{R,l}}{\partial r_i(t)}$ in $\mathbf{B}$ and $\mathbf{D}$ can be calculated as

$$\frac{\partial \bar{S}_{R,l}}{\partial r_i(t)} = -\frac{\partial w_l}{\partial r_i(t)} \tag{59}$$

$$\frac{\partial \tilde{S}_{R,l}}{\partial r_i(t)} = \left(1 - \sum_{n=1}^{N} p_{n,l}\right) \frac{\partial w_l}{\partial r_i(t)}. \tag{60}$$

Thus, according to the above analysis, we can calculate the matrix of first order derivative of the activation weight $w_l(t)$ to the attribute reliability $r_i(t)$ by

$$\frac{\partial w_l}{\partial r_i(t)} = \begin{bmatrix} \frac{\partial w_l}{\partial \alpha_l(t)} & \frac{\partial \mathbf{w_l}}{\partial \alpha_\mathbf{l}} \end{bmatrix} \begin{bmatrix} \frac{\partial \alpha_l(t)}{\partial r_i(t)} \\ \frac{\partial \alpha_\mathbf{l}}{\partial \mathbf{r_i}} \end{bmatrix} \tag{61}$$

with

$$\frac{\partial w_l}{\partial \alpha_k(t)} = \frac{\theta_l(t) \sum_{l=1}^{L} \theta_l(t)\alpha_l(t) - \theta_l^2(t)\alpha_l(t)}{\left(\sum_{l=1}^{L} \theta_l(t)\alpha_l(t)\right)^2} \tag{62}$$

$$\frac{\partial \mathbf{w_l}}{\partial \alpha_\mathbf{l}} = \begin{bmatrix} \frac{\partial w_l}{\partial \alpha_1(t)} & \cdots & \frac{\partial w_l}{\partial \alpha_L(t)} \end{bmatrix} \tag{63}$$

$$\frac{\partial \alpha_\mathbf{l}}{\partial \mathbf{r_i}} = \begin{bmatrix} \frac{\partial \alpha_1(l)}{\partial r_i(t)} & \cdots & \frac{\partial \alpha_L(l)}{\partial r_i(t)} \end{bmatrix}^{\mathrm{T}} \tag{64}$$

$$\frac{\partial w_l}{\partial \alpha_l(t)} = -\frac{\theta_l(t)\theta_l(t)\alpha_l(t)}{\left(\sum_{n=1}^{L} \theta_n(t)\alpha_n(t)\right)^2} \tag{65}$$

$$\frac{\partial \alpha_l(t)}{\partial r_i(t)} = \prod_{p=1}^{M} \left(\alpha_l^p(t)\right)^{J_p(t)} \ln\left(\alpha_l^i(t)\right) \frac{\partial \chi_i(t)}{\partial r_i(t)} \tag{66}$$

$$\frac{\partial \chi_i(t)}{\partial r_i(t)} = \frac{\bar{\delta}_i(t)}{\left(1 + \bar{\delta}_i(t) - r_i(t)\right)^2}. \tag{67}$$

Final, based the above derivation process of sensitivity analysis, the following corollary can be obtained.

Corollary 1: For the BRB-DR based OFTC framework as profiled by (3), suppose that there are M input attributes and N output degrees. The sensitivity coefficient of ith input attribute is expressed by the following equation:

$$u\left(r_i(t)\right) = \frac{1}{N} \sum_{n=1}^{N} \left| \frac{\partial p_n(t)}{\partial r_i(t)} \right| \tag{68}$$

where $S(r_i(t))$ is the sensitivity coefficient of the ith input attribute. $\frac{\partial p_n(t)}{\partial r_i(t)}$ can be obtained by (37)–(66). The sensitivity coefficient of the input attribute represents its effectiveness for the model output. For instance, if $u(r_1(t)) > u(r_2(t))$, the effectiveness of the first sensor is higher than the second sensor, and certain resources used to improve the reliability of the first sensor will have a better effect on the model output. Therefore, in practical engineering applications, the sensitivity coefficient can serve as a valuable reference for optimizing system design and maximizing the efficiency of resource allocation.

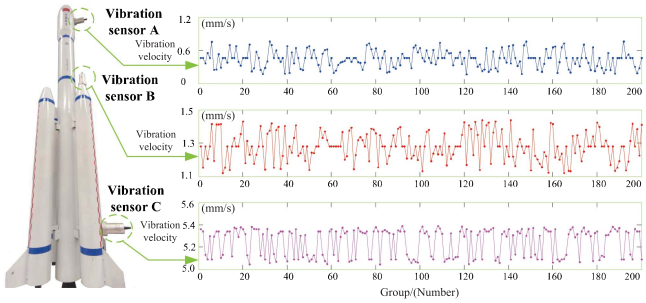


Fig. 2. Observation of vibration sensors

V. Case Study

In this section, in order to demonstrate the validity of the developed framework, a case study of the vibration sensor failure in a rocket is conducted.

A. Problem Formulation in the Sensor Network of Rocket

The observation data from the vibration sensor is employed in the navigation system of the rocket, and its accuracy directly affects its navigation accuracy. In FDOR of vibration sensor failure, there are three problems that needs to be solved.

1) *Uncertain expert knowledge:* During the preparation and launch phases of the rocket, the sensor works for a long time, and their failure is inevitable. On the other hand, the sensors of the same type are installed in different positions, and the disturbance factors of the sensor observation information in different positions are also different. Experts are unable to provide accurate sensor knowledge.
2) *Dynamic environment disturbance:* During the preparation and launch phases, the working environment of the rocket changes and the sensor observation information is influenced. Therefore, the noises in the observation information changes timely. These noise will reduce the reliability of the sensor observation information and influenced the performance of OFTC framework.
3) *High real-time requirement:* The rocket is a high-value system and the economic loss caused by failure is huge. As soon as a sensor fails, it needs to be diagnosed and handled in the shortest possible time. Thus, the OFTC framework should have high real-time performance.

In the section, the vibration sensor network is selected as the experiment objective and the experiment is carried out on the basis of hardware-in-the-loop simulation platform. The simulation rocket and the observation data of three vibration sensors (Sensor A, Sensor B, and Sensor C) are shown in Fig. 2. In the experiment, assume that the other parts of the navigation system are working properly, and do not effect the navigation accuracy of the rocket.

B. OFTC Framework of WSN

In the experiment, the time window's length ϖ is set to 50, and $T = 50$. When the number of observation data is less than

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TABLE I
REFERENCE POINTS AND VALUES OF SENSORS

Reference point	I_1	I_2	I_3	I_4
Sensor-A(S-A)	0	0.56	0.63	1.11
Sensor-B(S-B)	1.28	1.35	1.37	1.42
Sensor-C(S-C)	5.13	5.35	5.42	5.55

T, the dynamic reliability of sensor is set as the mean reliability of its experiment process.

Then, the observation data of three sensors can be analyzed. As shown in Table I, the reference points of three sensors are determined by experts, where I_1, I_2, I_3 and I_4 respectively stand for the small, medium, slightly large and large vibration. It need to be pointed out that the reference points directly influence the accuracy of the FDOR, and they should follow the two principles: cover all data changes and the distance between the reference points is appropriate. The inputs to the fault diagnosis model are observations from three sensors, and the inputs to the faulty sensor output reconstruction model are observations from the remaining two available sensors. In the fault diagnosis model, the standard fault variable is set as $\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$. The reference points of reconstruction model are same as the reference points of the failure sensor in Table I.

According to the reference points, the initial fault diagnosis model of the OFTC framework can be developed by experts, where the initial weights of the belief rules are assumed to be one. In the experiment, three output reconstruction models are developed and the weights of the attributes are assumed to be equally important.

C. Training and Testing

In the experiment, the FDOR model are optimized sequentially. Optimization iterations of the optimization model of these models are set as 100 and 200, respectively. A total of 800 observations are collected from the three sensors. 400 observations are randomly selected as training data, and the remaining 400 observations are used as test data. It should be noted that to ensure the interpretability of the OFTC framework, the parameters in the FDOR model for failure sensors should satisfy the constraints as shown in (35)–(35). After the training part of the OFTC framework, the optimized FDOR model for three sensors are shown in Tables II–IV.

The testing output of the optimized fault diagnosis model can be shown in Fig. 3. In Fig. 3, the horizontal coordinate indicates the 400 sets of training data used in the experiment, and each of these four hundred sets of training data corresponds to the four types of failure modes from o1 to o4 in turn. If the fault diagnosis is completely accurate, the output results (blue "$*$") of each dataset should completely coincide with the red line. Therefore, all results that do not coincide with the red line are misdiagnosed. Based on this, the fault diagnosis accuracy can be obtained as 98.50% . This shows that the optimized fault diagnosis model is able to diagnose the failure sensor accurately. 50 experiments has been conducted on the fault diagnosis model and the mean and variance are 97.44% and 4.1911×10^{-3}, respectively, which well demonstrated the stability after optimization.

According to the fault diagnosis model, the failure sensor can be determined, and its observation data can be reconstructed

TABLE II
OPTIMIZED OUTPUT RECONSTRUCTION MODEL OF S-A

No.	Sensor B	Sensor C	Rule weight	Output Confidence $\{O_4, O_3, O_2, O_1, \Theta\}$
1	I_4	I_4	0.1168	{0.2567 0.0977 0.1764 0.3986 0.0705}
2	I_4	I_3	0.5005	{0.3366 0.2704 0.0630 0.3153 0.0147}
3	I_4	I_2	0.2619	{0.1177 0.2178 0.1360 0.4145 0.1140}
4	I_4	I_1	0.0962	{0.1401 0.1294 0.3281 0.2845 0.1179}
5	I_3	I_4	0.5790	{0.2594 0.4961 0.1274 0.1149 0.0022}
6	I_3	I_3	0.9631	{0.2098 0.3916 0.1175 0.2571 0.0239}
7	I_3	I_2	0.7651	{0.0496 0.3821 0.1094 0.4254 0.0336}
8	I_3	I_1	0.3983	{0.4282 0.0457 0.1410 0.2402 0.1449}
9	I_2	I_4	0.0259	{0.3342 0.0502 0.0865 0.3650 0.1641}
10	I_2	I_3	0.6755	{0.1070 0.3820 0.2080 0.2777 0.0253}
11	I_2	I_2	0.3073	{0.0749 0.5720 0.1398 0.0208 0.1924}
12	I_2	I_1	0.3344	{0.1407 0.3925 0.0000 0.3494 0.1185}
13	I_1	I_4	0.6044	{0.3027 0.1281 0.3069 0.2523 0.0101}
14	I_1	I_3	0.7035	{0.0944 0.3265 0.3681 0.1913 0.0196}
15	I_1	I_2	0.7026	{0.2822 0.1860 0.2144 0.1262 0.1912}
16	I_1	I_1	0.6876	{0.2330 0.1784 0.2302 0.3293 0.0292}

TABLE III
OPTIMIZED OUTPUT RECONSTRUCTION MODEL OF S-B

No.	Sensor A	Sensor C	Rule weight	Output Confidence $\{O_4, O_3, O_2, O_1, \Theta\}$
1	I_4	I_4	0.6255	{0.0144 0.1502 0.1504 0.5858 0.0993}
2	I_4	I_3	0.0006	{0.1564 0.1060 0.3310 0.3327 0.0738}
3	I_4	I_2	0.6945	{0.5676 0.1664 0.0446 0.0284 0.1930}
4	I_4	I_1	0.4170	{0.2683 0.0328 0.4570 0.2300 0.0119}
5	I_3	I_4	0.7366	{0.5614 0.1683 0.0582 0.0218 0.1902}
6	I_3	I_3	0.4567	{0.4089 0.1754 0.0736 0.1470 0.1951}
7	I_3	I_2	0.2679	{0.1090 0.2887 0.0107 0.5762 0.0153}
8	I_3	I_1	0.7574	{0.3322 0.1526 0.0376 0.3252 0.1523}
9	I_2	I_4	0.4713	{0.6554 0.0438 0.0893 0.1082 0.1033}
10	I_2	I_3	0.4348	{0.6494 0.0380 0.1434 0.0140 0.1553}
11	I_2	I_2	0.2803	{0.4102 0.0495 0.3079 0.0679 0.1645}
12	I_2	I_1	0.3952	{0.2059 0.0686 0.1585 0.4033 0.1637}
13	I_1	I_4	0.3072	{0.4927 0.1784 0.0051 0.2347 0.0891}
14	I_1	I_3	0.7665	{0.0914 0.1840 0.5927 0.1006 0.0313}
15	I_1	I_2	0.2508	{0.3873 0.0931 0.2109 0.2292 0.0795}
16	I_1	I_1	0.3663	{0.2181 0.4175 0.1666 0.0320 0.1658}

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TABLE IV
OPTIMIZED OUTPUT RECONSTRUCTION MODEL OF S-C

No.	Sensor		Rule	Output Confidence
	A	B	weight	$\{O_4, O_3, O_2, O_1, \Theta\}$
1	I_4	I_4	0.0871	{0.2084 0.3309 0.0604 0.2847 0.1155}
2	I_4	I_3	0.7259	{0.0557 0.3241 0.0875 0.5060 0.0267}
3	I_4	I_2	0.5554	{0.2987 0.0710 0.1720 0.4376 0.0207}
4	I_4	I_1	0.8024	{0.0069 0.5239 0.2680 0.1918 0.0095}
5	I_3	I_4	0.7247	{0.1243 0.3025 0.0639 0.4876 0.0216}
6	I_3	I_3	0.9111	{0.1735 0.4769 0.2215 0.1179 0.0102}
7	I_3	I_2	0.1300	{0.1626 0.0352 0.1756 0.5492 0.0775}
8	I_3	I_1	0.9293	{0.1365 0.4353 0.2300 0.1875 0.0107}
9	I_2	I_4	0.4708	{0.2212 0.0525 0.2912 0.2632 0.1718}
10	I_2	I_3	0.2544	{0.0450 0.1404 0.4155 0.2787 0.1204}
11	I_2	I_2	0.9885	{0.1481 0.0550 0.2758 0.3443 0.1768}
12	I_2	I_1	0.1006	{0.1397 0.1542 0.0852 0.4461 0.1748}
13	I_1	I_4	0.5098	{0.0435 0.5931 0.1478 0.1849 0.0307}
14	I_1	I_3	0.5830	{0.5790 0.1181 0.1463 0.0750 0.0816}
15	I_1	I_2	0.0591	{0.1552 0.0613 0.1405 0.4835 0.1594}
16	I_1	I_1	0.0586	{0.1836 0.3335 0.2821 0.0817 0.1191}

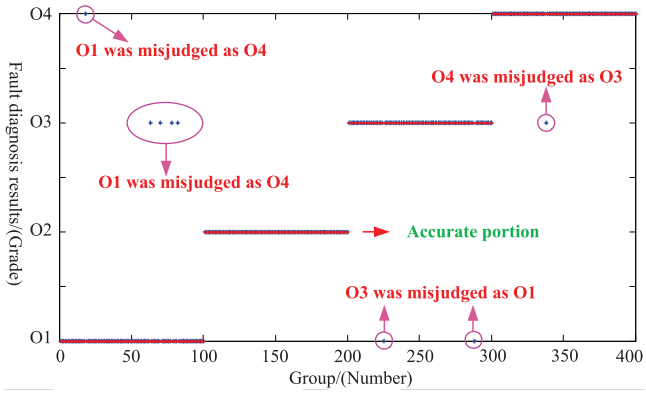


Fig. 3. Fault diagnosis results.

on the basis of the available sensors. Then, the reconstructed observation data of the failure sensor and the observation data of the normal sensor are used in the navigation system. Evidently in Fig. 4, sensor failure has a greater influence on the rocket's accuracy, and the framework has the ability to improve the navigation accuracy obviously when the vibration sensor fails. Noted that the different sensor failures have different influences on the navigation accuracy of the rocket. Hence, it can be concluded that the OFTC framework can effectively solve the sensor failure with environment disturbance.

In order to demonstrate the validity of the developed framework, comparative experiments are carried out with the knowledge-based method, the data based method and current BRB models. In the OFTC framework, its fault diagnosis model is the first part and its accuracy directly affects the framework performance. Therefore, in this section, the traditional BRB model, the Fuzzy theory, the Neural network, the BRB model

TABLE V
COMPARATIVE EXPERIMENT RESULT

Number	Method	Accuracy	Stability
1	Initial BRB-DR	69.28%	1.426
2	Optimized BRB-DR	98.50%	0.085
3	Fuzzy theory	83.50%	0.153
4	Neural network	80.20%	0.658
5	He *et al.* [38]	84.54%	1.102
6	Zhou *et al.* [39]	86.57%	0.461

developed by He et al. [38] and Zhou et al. [39] are selected as the comparative methods to diagnose the sensor failure. The comparative experiments result is shown in Table V. Compared with the neural network and fuzzy theory, the diagnosis accuracy increases 15.00% and 18.30%, respectively. It becomes apparent that the fuzzy theory and neural network cannot diagnose the sensor failure accurately. That is because the the uncertainty expert knowledge and lack of observations. What is more, compared with the BRB models of He et al. [38] and Zhou et al. [39], the accuracy of BRB-DR improves 13.96% and 11.93%, respectively. We can summarize that the BRB-DR model can address the environment influence and its modeling ability is improved. Moreover, as compared to the original BRB-DR model, the diagnosis accuracy improves 29.22% after optimization which demonstrates the validity of the developed optimization model. Thus, in the experiment, it can be summarized that the performance of the BRB-DR based OFTC framework is better than others. In terms of model stability analysis, the article records the variance of each method after conducting thirty experiments, as shown in Table V. Among these methods, the BRB-DR method exhibits the smallest variance, which is only 0.085. This indicates that the fault diagnosis model constructed using the BRB-DR method has high stability compared to other methods.

D. Sensitivity Analysis

In order to analyze the validity of the dynamic reliability of three sensors, the sensitivity analysis is conducted based on the optimized fault diagnosis model and optimized output reconstruction models of the three sensors.

The optimized fault diagnosis model and the optimized output reconstruction models of three sensors are indicated in Tables II–V. Then from Fig. 5, the sensitivity coefficients for model input reliability can be derived. As can be seen in Fig. 5, the reliability of the observed information from Sensor B is higher compared to the other two sensors. That is to say, in the developed OFTC framework, the Sensor B has the greatest impact. Thus, in the case of limited resources, improving the performance of the Sensor B will give an improvement the accuracy of the fault diagnosis and the effect of the fault tolerant control more significantly.

VI. CONCLUSION

In this article, a new OFTC framework is developed to address the integration of sensor faults with unreliable information from

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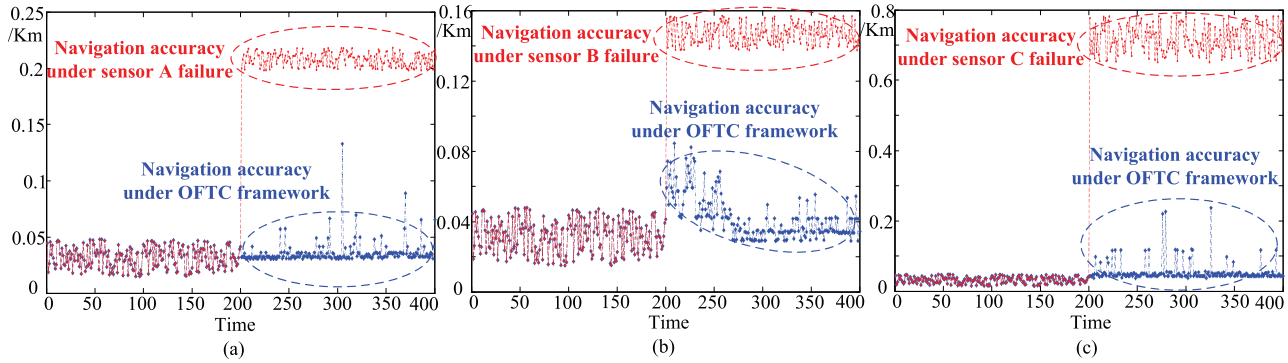


Fig. 4. Navigation accuracy of the rocket under sensor failure and the OFTC framework. (a), (b), and (c) denote the deviation of navigation accuracy when sensors A, B, and C fail, respectively.

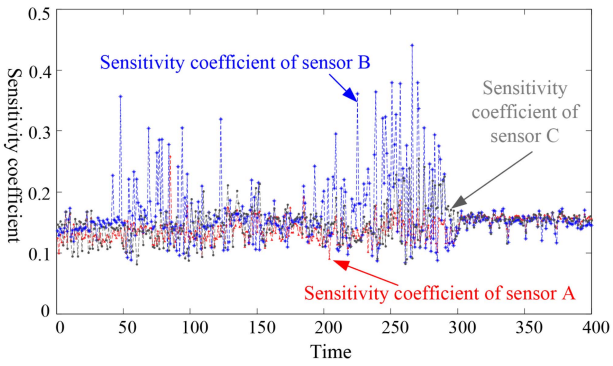


Fig. 5. Sensors sensitivity coefficients.

multiple sources. This OFTC framework aims to solve three problems in sensor fault diagnosis and tolerant control: Lack of observation information, uncertainty expert knowledge, and dynamic environment disturbance. Responding to these problems, this article mainly makes three contributions.

First of all, a new BRB-DR is developed, where the influence of the environment interference is addressed. In BRB-DR, a new calculation method of dynamic reliability of attribute is proposed according to the segment statistics strategy. Second, to improve the reliability of complex systems under sensor failure, an OFTC framework is constructed based on the BRB-DR. Under new framework, the nearest neighbor strategy and the output reconstruction method are applied for the failure sensor. What is more, a sensitivity analysis about the dynamic reliability to the model output is established. The obtained sensitivity coefficient of the dynamic attribute reliability can quantitatively represent its effectiveness to the model output that can provide decision making support for improving system design.

The developed OFTC framework uses the output reconstruction strategy in the fault tolerant stage. The observation information of the failure sensor is constructed according to the correlation of the sensor observation information. However, the extent to which correlations between sensors can or cannot be used in the framework needs to be investigated. In addition, the influence of factors such as environmental disturbances can affect the diagnostic accuracy of the model in practical engineering applications, and these issues pose a challenge for future work.

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